Scaling Laws for Dense Retrieval: a scaled-down illustration, with an audit

Companion report for the SIGIR '24 paper "Scaling Laws For Dense Retrieval" (Fang, Zhan, Ai, Mao, Su, Chen, Liu; arXiv 2403.18684v2). All numbers in this report are generated by script from `data/results/*.json`*,* `data/results.csv` *and* `data/results_summary.csv`*.*

0. Audit result box

Group	Verdict
A. Course brief	PASS WITH CAVEATS
B. First assignment	PASS WITH CAVEATS
C. Fidelity to the paper	PASS WITH CAVEATS: definitions match; training length does not
D. Experimental hygiene	PASS WITH CAVEATS

Compliance matrix: 42 items = 24 MATCH, 12 DEVIATION-JUSTIFIED, 3 DEVIATION-IMPACTFUL, 3 UNVERIFIED (full table: `AUDIT_COMPLIANCE_MATRIX.md`).

- **DEVIATION-IMPACTFUL:** B8 (results.csv, SUMMARY.md, CHANGES.md); C7 (Optimisation length); C9 (Metric: contrastive entropy = NLL of the positive among 1+256 candidates).
- **UNVERIFIED:** A4 (Group of 2–3 students); C18 (Mixed precision); D6 (Train loss vs eval loss comparable (to discuss over-fitting)?).

Use this wording: "in a heavily scaled-down setting, loss moves in the same direction as in the paper". **Do not write** that the paper was reproduced.

1. Executive summary

Paper says (Sec. 1, Sec. 4.1–4.2, Fig. 3–4): the test loss of dense retrieval models, measured by a continuous "contrastive entropy", follows a power law in model size and in the number of annotated pairs.

Our experiment shows (Sec. 6 below): with 5 BERT sizes (N from 512,128 to 86,235,648 non-embedding parameters), D = 10,000 training pairs and 3 seeds, the mean loss falls from 0.7265 to 0.1581 nats. With BERT-Small and D from 2,500 to 40,000 pairs it falls from 0.3658 to 0.1596. A 2-parameter power law fits both sweeps well (R^2 on the log scale 0.986 and 0.989; exponents 0.299 and 0.285) and the exponents are stable when one point is left out (0.259–0.320 and 0.267–0.294).

Caveats that matter: (1) our runs last 60–939 optimizer steps, the paper fixes 10,000 (Sec. 3.4); (2) the 3-parameter exponent of the data sweep is unstable (0.260–0.770 under leave-one-out); (3) two adjacent pairs (Small vs Medium, D = 10,000 vs 20,000) are not separated by the seed-to-seed spread; (4) the metric is a negative log-likelihood in nats, **not** a perplexity: the "Contrastive Perplexity" [OLD-LABEL] axis label in the original plots was wrong and is corrected here; random scoring would give ln(257) = 5.549.

2. Background for newcomers

Information retrieval (IR) is finding the documents (here: short text passages) that answer a query. **Dense retrieval** does this with neural networks: an *encoder* turns every query and every passage into a list of numbers (a vector, also called *embedding*). A passage is judged relevant when its vector points in nearly the same direction as the query vector; the score is the dot product. Because passage vectors can be computed once in advance, search over millions of passages is fast.

Example. Query "capital of California" and passage "Sacramento is the capital of California" should get a high score; a passage about football should get a low one (Figure F2).

Training with a contrastive loss. The model sees a query with one labelled relevant passage (the *positive*) and many random wrong passages (*negatives*). The scores of all candidates are turned into probabilities with a softmax. The loss is the negative logarithm of the probability given to the positive: it is 0 if the model puts all probability on the positive and large if it does not. This is the *negative log-likelihood (NLL)*, measured in *nats* (natural logarithm). With K candidates, a model that guesses at random gets ln(K); here K = 1 + 256 = 257, so ln(K) = 5.549 (Figure F3, F10).

Scaling law. In many neural networks the loss falls as a smooth power law, $L \approx a \cdot X^{(-\alpha)}$, when the model size N or the data size D grows. On log–log axes this is a straight line whose slope is $-\alpha$. If it holds, small, cheap experiments can predict large ones, which is why the topic matters; the paper argues it also helps allocate a budget between annotation and compute (Sec. 5).

3. What the paper says

(Every statement cites the section, figure, table or equation of arXiv 2403.18684v2. Metadata verified against Crossref and the arXiv API: SIGIR '24, Washington DC, July 14–18 2024, pp. 1339–1349, DOI 10.1145/3626772.3657743.)

Paper says: the research questions are how model size, annotated-data size and annotation quality affect dense retrieval (Sec. 1, Sec. 4 opening).

Paper says: the encoder is a pre-trained Transformer with a projection layer, shared by queries and passages, scored by inner product (Sec. 3.1–3.2, Eq. 1–2, 4); training uses random and in-batch negatives, 256 negatives per step, a fixed 10,000 steps (Sec. 3.4, Eq. 3). Learning rate, batch size and temperature are not stated in the text (UNVERIFIED in the paper).

Paper says: the evaluation metric is the "contrastive entropy", the NLL of Eq. 5 with 256 random negatives per test pair; a 100,000-passage sub-corpus is used in the metric study; it correlates with MAP@10, NDCG@10 and Recall@1000 (Sec. 3.5, Fig. 2), with a possible "emergent" jump near 0.25.

Paper says: N is the number of non-embedding parameters, from 0.5M (BERT-Tiny) to 82M (BERT-Base) (Sec. 3.2 and its footnote 2; Fig. 6 legend); the Introduction says "0.5 to 87 million" (Sec. 1), an internal inconsistency we could not resolve. D is the number of annotated query–passage pairs (Sec. 3.1, 4.2).

Paper says: model-size law L(N) = (A/N)^α + δ (Eq. 6) with, for MS MARCO, A = 3.22e4, α = 0.53, δ = 0.04, R^2 = 0.991 (Table 1, Fig. 3). Data-size law L(D) = (B/D)^β + δ (Eq. 7) with B = 3.49e3, β = 1.05, δ = 0.05, R^2 = 0.954 (Table 2, Fig. 4), fitted on BERT-Base with up to all annotations. A joint law (Eq. 8) has A ≈ 3.6e4, B ≈ 7.1e3, α ≈ 0.56, β ≈ 1.31, δ ≈ 0.03 (Sec. 4.4, Eq. 9–10, Fig. 6). The constant δ is read as an irreducible loss (Sec. 4.1).

Paper says: the scaling holds for human labels and for three kinds of generated labels (ICT, docT5query, ChatGLM3); ICT has the smallest slope, and the Fig. 5 caption says ChatGLM3 has the steepest slope and passes human labels at 500K while the text says human labels give the best models (Sec. 4.3, Fig. 5). With the fitted joint law and a cost model, a $20,000 budget favours a model of about 13 billion parameters if inference cost is ignored, and million-scale models if it is included (Sec. 5, Fig. 7–8).

Paper says (limitations, Sec. 6): in-domain evaluation only, no hard negatives or distillation, single-vector retrievers only. The arXiv listing notes that v2 "fixes a bug in the experiments".

4. Our experiment

Goal. An inexpensive illustration of the two basic trends (loss vs N, loss vs D) with an honest error estimate. It is **not** a reproduction.

Aspect	Paper (section / figure)	Our experiment
Task / data	MS MARCO passage (8.8M passages, 0.5M train queries) and T2Ranking (Sec. 3.3)	MS MARCO only; passage file truncated to 8,326,970 passages (94.18%)
Models	24 Google BERT checkpoints 0.5M–82M non-emb. params; ERNIE for T2Ranking (Sec. 3.2)	5 BERT checkpoints: Tiny, Mini, Small, Medium, Base (same family as the paper's red stars, Fig. 3)
N definition	non-embedding parameters (Sec. 3.2, footnote 2)	non-embedding incl. projection layer and unused BERT pooler (= the paper's Fig. 6 legend sizes when divided by 2^20)
D definition	number of annotated query–passage pairs (Sec. 3.1, 4.2)	number of (qid, pid) training pairs
Architecture	shared encoder + projection to 768; CLS or mean pooling (Sec. 3.2, Eq. 2)	shared BERT + mean pooling + Linear(H→768); no normalisation
Training loss	softmax cross-entropy, random + in-batch negatives (Sec. 3.4, Eq. 3)	same: 1 positive + 256 random negatives shared by the batch; mean CE; no temperature
Optimisation length	fixed 10,000 steps (Sec. 3.4)	3 epochs → 60–939 steps (depends on D); 237 steps for the model sweep
lr / batch / optimiser	not stated in the paper (UNVERIFIED)	lr 1e-4, batch 128, AdamW, linear decay, warmup 10 steps, bf16
Metric	contrastive entropy = −log softmax of the positive among 1+256 candidates (Sec. 3.5, Eq. 5)	same quantity (mean NLL, nats), K = 257
Evaluation set	dev queries (~7,000, Sec. 3.3); 100,000-passage sub-corpus used for the metric study (Sec. 3.5)	first 1,000 dev.small queries by qid (1,086 pairs); 100,000-passage pool; 5 repeats; fixed eval seed 1
Checkpoint reported	best test loss during training, no early stopping (Sec. 4.1)	final checkpoint after epoch 3
Model sweep	full training set, many models (Sec. 4.1, Fig. 3)	D = 10,000 pairs, 5 models
Data sweep	BERT-Base, up to all ~500K annotations (Sec. 4.2, Fig. 4)	BERT-Small, D = 2,500 … 40,000
Fit form	L = (A/N)^α + δ (Eq. 6); L = (B/D)^β + δ (Eq. 7); joint Eq. 8	3-param a·X^-α + c (same family; a = A^α) and a 2-param a·X^-α (extra, main)
Seeds	not stated (UNVERIFIED)	3 training seeds 42/43/44; evaluation seed fixed
Annotation quality, budget allocation	Sec. 4.3 (ICT, docT5query, ChatGLM3), Sec. 5	not reproduced

Data. MS MARCO passage ranking. The corpus file has 8,326,970 passages (94.18% of the full collection; the file is cut at the end, ids 0…8,326,969). Of 532,751 unique training pairs, 3,311 (0.62%) point to missing passages and are dropped, leaving 529,440 pairs over 499,872 queries. D pairs are the first D of a seeded shuffle.

Models. Five Google BERT miniatures, parameters in Table A1 of the appendix; one shared encoder, mean pooling, Linear(H→768), dot-product score, no normalisation, no temperature (Figure F2).

Training. 3 epochs, lr 1e-4, batch 128, AdamW (weight decay 0.001, β2 0.999), linear decay, 10 warm-up steps,

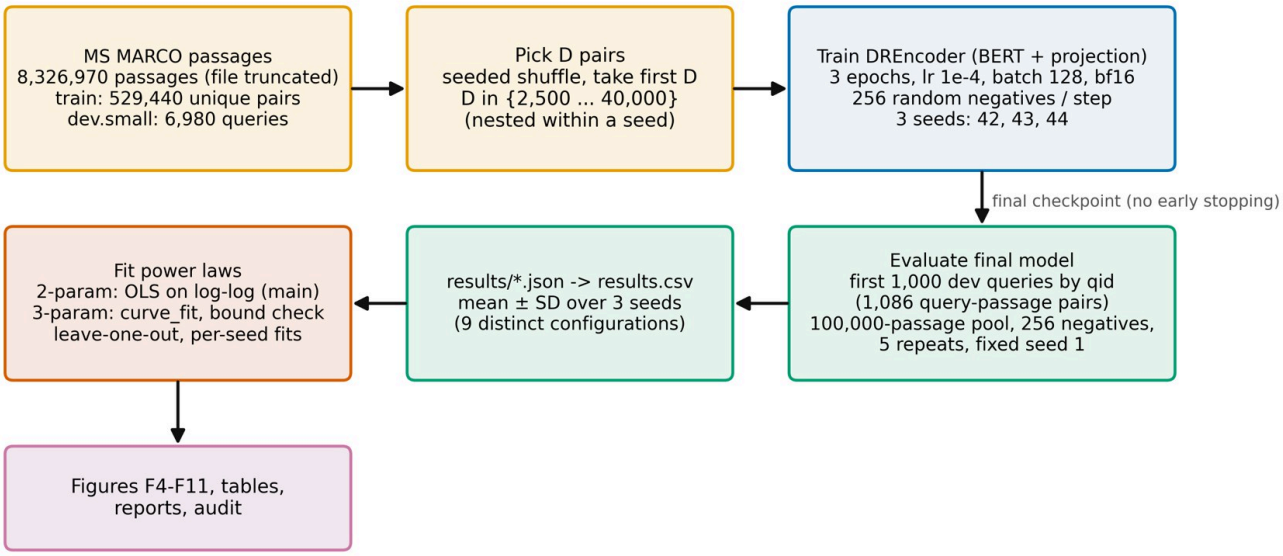
gradient clipping 1.0, bf16, gradient checkpointing. Per step, 256 random corpus passages are shared by the batch as negatives (K = 257). Settings are identical for all 27 runs (read from each run's `training_args.bin`).

Evaluation. The 1,000 dev.small queries with the smallest qids (1,086 query–positive pairs); a fixed pool of 100,000 passages drawn from the 8,325,884 passages that are not positives of those queries; for each pair, 256 negatives drawn from the pool, 5 repeats (5,430 draws per model); evaluation seed 1 for every model; the mean over pairs is the reported loss.

Seeds. Training seeds 42, 43, 44 change the projection initialisation, the data subset and order and the training negatives; the evaluation seed does not change.

Hardware and time. One NVIDIA RTX 5880 Ada Generation (49,140 MiB; driver 580.178.04, nvidia-smi on 2026-10-29); Intel Core Ultra 9 285K (24 logical CPUs), 93 GiB RAM; PyTorch 2.11.0+cu128, Transformers 4.52.4; peak GPU memory (allocated) 4645.6 MiB. Summed over the 27 runs: training 1966 s, evaluation 1368 s, total 3495 s (58.2 min). The last orchestrator session (seeds 43/44 and new configurations) lasted 51.1 min.

F1. Experiment workflow (scaled-down)



5. Audit results

5.1 Compliance summary

Verdicts and counts are in the box of Section 0; the complete matrix with evidence is `AUDIT_COMPLIANCE_MATRIX.md`. Group C is the one that limits what may be claimed.

5.2 Findings about the code

Code does: the architecture in `modeling.py` is unchanged from the original repository (the file is byte-identical). Mean pooling, Linear(H→768), plain dot product, no temperature, cross-entropy over 1 positive and 256 shared random negatives, mean over the batch (`modeling.py` lines 47–72, 112–118).

Code does: `contrastive_perplexity` computes, per (query, positive) pair, the cross-entropy over 1 + 256 candidates and averages the pair losses and the 5 repeats. There is no `exp`, so the value is an NLL in nats, not a perplexity. Chance level is ln(257) = 5.549; our worst model has 0.131 of this and our best 0.028. For reference only, exp of the mean is 2.068 for the worst model and 1.171 for the best.

Code does: training and evaluation both use K = 257 and the same formula, but the training loss in the logs is a running mean over the whole run, computed in train mode (dropout) with bf16 on pairs already seen. The final dev loss is computed on the final model in eval mode with fp32 on held-out queries. They are **not comparable**; the training curves (F8) show the training loss still falling at the end, and the last-window training loss for BERT-Small (0.0555) is far below its dev loss (0.2345), but this cannot be called over-fitting without a like-for-like measurement (for example evaluating the final model in eval mode on held-out training queries, not done here).

Code does: changes from the original repository are limited to paths, robust file reading, a missing-argument fix, checkpoint saving, optional pre-loaded data, a 1,000-query evaluation cap and a multi-seed orchestrator (see `CHANGES.md`). None of them alters the loss, the model or the metric definition; the cap on queries and the pool size are intentional scale-downs.

Our audit shows: no query leaks from training into evaluation (train qids ∩ dev qids = 0, ∩ the evaluation qids = 0, identical query texts = 0). D subsets are nested within a seed and independent across seeds (10,000-pair overlaps 203–207 against 189 expected by chance). The evaluation pool is identical for every model (rebuild

check, sha1 prefix f965e175b62f). The 1,000-query evaluation subset is a bit unrepresentative: mean query length 5.30 words versus 6.03 for the other dev queries.

Our audit shows: the missing 5.8% of the corpus are the highest passage ids. Lost training pairs belong to slightly shorter queries (5.74 vs 5.97 words); dev loses nothing. Impact is judged low, though whether ids carry any topical order is UNVERIFIED. The training pair file has 10 duplicated lines (532,761 lines, 532,751 unique pairs).

Our audit shows: the non-embedding count used for N includes the unused BERT pooler and the projection layer. In units of 2^20 the five sizes are 0.5, 3.3, 12.7, 24.7, 82.2 M, exactly the labels in the paper's Fig. 6 legend, so the counting convention agrees with the paper. Our x-axes use 10^6 units.

5.3 Independent recomputation (from the 27 JSON files)

Means and standard deviations (SD, ddof = 1) come from the JSON files, which agree with `results.csv` (max difference 0.0e+00) and `results_summary.csv` (rounding only). There are 9 distinct configurations (not 10: BERT-Small with D = 10,000 belongs to both sweeps). Fits: all six SUMMARY values reproduce to the printed precision.

Sweep	Fit	a	exponent	c	R^2 (log)	R^2 (linear)	notes
Model size N	2-param L = a·X^-α (OLS in log–log)	34.79	0.299	—	0.9858	0.9867	main fit
Model size N	3-param L = a·X^-α + c (curve_fit)	206.29	0.440	0.093	—	0.9991	bound flags: none; 5 points, 3 parameters
Data size D	2-param L = a·X^-α (OLS in log–log)	3.32	0.285	—	0.9887	0.9882	main fit
Data size D	3-param L = a·X^-α + c (curve_fit)	8.97	0.444	0.084	—	0.9918	bound flags: none; 5 points, 3 parameters

The 3-parameter fits reach the same optimum from all 27 starting points (model) and 27 (data), with or without bounds, and touch no bound. **However** they have 2 degrees of freedom: leave-one-out gives 3-parameter exponents of 0.426–0.492 (model) and 0.260–0.770 (data, with 1 fit hitting the bound c ≥ 0); per-seed fits give 0.367–0.489 and 0.247–0.669. The 2-parameter exponents are much steadier: leave-one-out 0.259–0.320 and 0.267–0.294, per seed 0.292–0.307 and 0.258–0.310 (Figure F9, Tables A2–A3). Counting parameters without the output layer gives a model exponent of 0.288 instead of 0.299.

Adjacent points (description only; n = 3 per point):

Sweep	From	To	Mean ± SD (from)	Mean ± SD (to)	Difference	±SD intervals overlap	Seed min–max ranges overlap	Seeds (of 3) where larger setting is better
model	BERT-Tiny D=10000	BERT-Mini D=10000	0.7265 ± 0.0145	0.3707 ± 0.0220	0.3558	no	no	3
model	BERT-Mini D=10000	BERT-Small D=10000	0.3707 ± 0.0220	0.2345 ± 0.0129	0.1362	no	no	3
model	BERT-Small D=10000	BERT-Medium D=10000	0.2345 ± 0.0129	0.2149 ± 0.0155	0.0196	yes	yes	3
model	BERT-Medium D=10000	BERT-Base D=10000	0.2149 ± 0.0155	0.1581 ± 0.0177	0.0568	no	no	3
data	BERT-Small D=2500	BERT-Small D=5000	0.3658 ± 0.0187	0.2830 ± 0.0063	0.0828	no	no	3
data	BERT-Small D=5000	BERT-Small D=10000	0.2830 ± 0.0063	0.2345 ± 0.0129	0.0485	no	no	3
data	BERT-Small D=10000	BERT-Small D=20000	0.2345 ± 0.0129	0.2059 ± 0.0217	0.0286	yes	yes	3
data	BERT-Small D=20000	BERT-Small D=40000	0.2059 ± 0.0217	0.1596 ± 0.0069	0.0463	no	no	3

Of 8 adjacent pairs, 2 have overlapping ±SD intervals: Small vs Medium (difference 0.0196) and D = 10,000 vs 20,000 (difference 0.0286). The larger setting is better in all 3 seeds for every pair, but with n = 3 this is only descriptive.

Comparison with the paper's exponents (the quantities are defined by the same formula family; the regimes are not the same):

Quantity	Paper (MS MARCO; Tables 1-2)	Ours	Comparable?
Model-size exponent α (3-param family; Table 1, Sec. 4.1)	0.53	0.440	No: see caveats
Model-size constant δ_N / c (Table 1)	0.04	0.093	No: see caveats
Model-size fit R^2 (Table 1)	0.991	0.9991	Descriptive only (5 points, 3 parameters)
Data-size exponent β (3-param family; Table 2, Sec. 4.2)	1.05	0.444	No: see caveats
Data-size constant δ_D / c (Table 2)	0.05	0.084	No: see caveats
Data-size fit R^2 (Table 2)	0.954	0.9918	Descriptive only
2-param exponents (no constant)	not reported	0.299 (model), 0.285 (data)	No: the paper has no such fit

Why it is not a like-for-like comparison: (i) 237 steps against 10,000 (Sec. 3.4); (ii) D from 2,500 to 40,000 on BERT-Small against up to about 500K on BERT-Base (Sec. 4.2, Fig. 4); (iii) final checkpoint against best test loss (Sec. 4.1); (iv) smaller evaluation set; (v) fits on 5 points with 3 free parameters. The exponents are lower than

the paper's, and both our constants (0.093, 0.084) are larger than the paper's (0.04, 0.05). We draw no conclusion from these differences.

6. Experimental results

Sweep	Model	N (non-emb., incl. fc)	D (pairs)	Steps	Seeds	Mean ± SD (nats)	Seed 42 / 43 / 44	exp(mean) (supplementary)	Train time (s)	Eval time (s)
model sweep	BERT-Tiny	512,128	10,000	237	3	0.7265 ± 0.0145	0.7424 / 0.7232 / 0.7140	2.068	21.3	17.3
model sweep	BERT-Mini	3,422,208	10,000	237	3	0.3707 ± 0.0220	0.3759 / 0.3466 / 0.3896	1.449	34.7	22.5
model sweep	BERT-Small	13,266,176	10,000	237	3	0.2345 ± 0.0129	0.2281 / 0.2493 / 0.2261	1.264	41.4	32.0
model sweep	BERT-Medium	25,875,712	10,000	237	3	0.2149 ± 0.0155	0.1977 / 0.2278 / 0.2191	1.240	75.8	60.2
model sweep	BERT-Base	86,235,648	10,000	237	3	0.1581 ± 0.0177	0.1785 / 0.1466 / 0.1492	1.171	201.2	193.7
data sweep	BERT-Small	13,266,176	2,500	60	3	0.3658 ± 0.0187	0.3618 / 0.3494 / 0.3862	1.442	11.2	31.6
data sweep	BERT-Small	13,266,176	5,000	120	3	0.2830 ± 0.0063	0.2831 / 0.2892 / 0.2766	1.327	21.4	32.0
data sweep	BERT-Small	13,266,176	10,000	237	3	0.2345 ± 0.0129	0.2281 / 0.2493 / 0.2261	1.264	41.4	32.0
data sweep	BERT-Small	13,266,176	20,000	471	3	0.2059 ± 0.0217	0.1834 / 0.2266 / 0.2078	1.229	82.8	33.4
data sweep	BERT-Small	13,266,176	40,000	939	3	0.1596 ± 0.0069	0.1656 / 0.1612 / 0.1520	1.173	165.6	33.4

Table: mean ± SD over 3 seeds of the contrastive entropy (NLL, nats, K = 257, lower is better). `exp(mean)` is supplementary only. The row for BERT-Small with D = 10,000 appears in both sweeps; it is the same 3 runs.

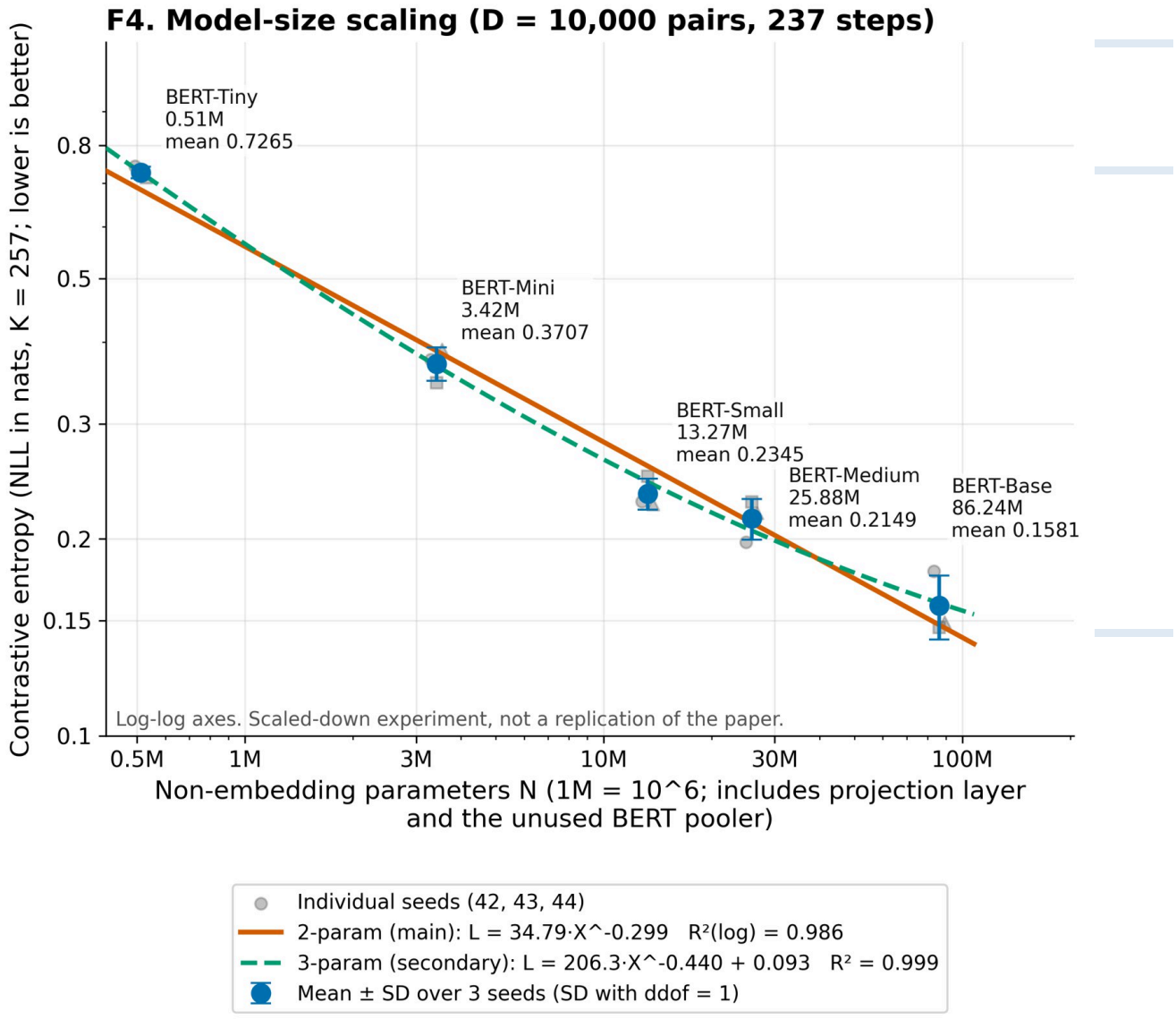


Figure F4. Loss against non-embedding parameters (log–log). Blue: mean ± SD over 3 seeds; grey: single seeds; solid: 2-parameter fit (main); dashed: 3-parameter fit.

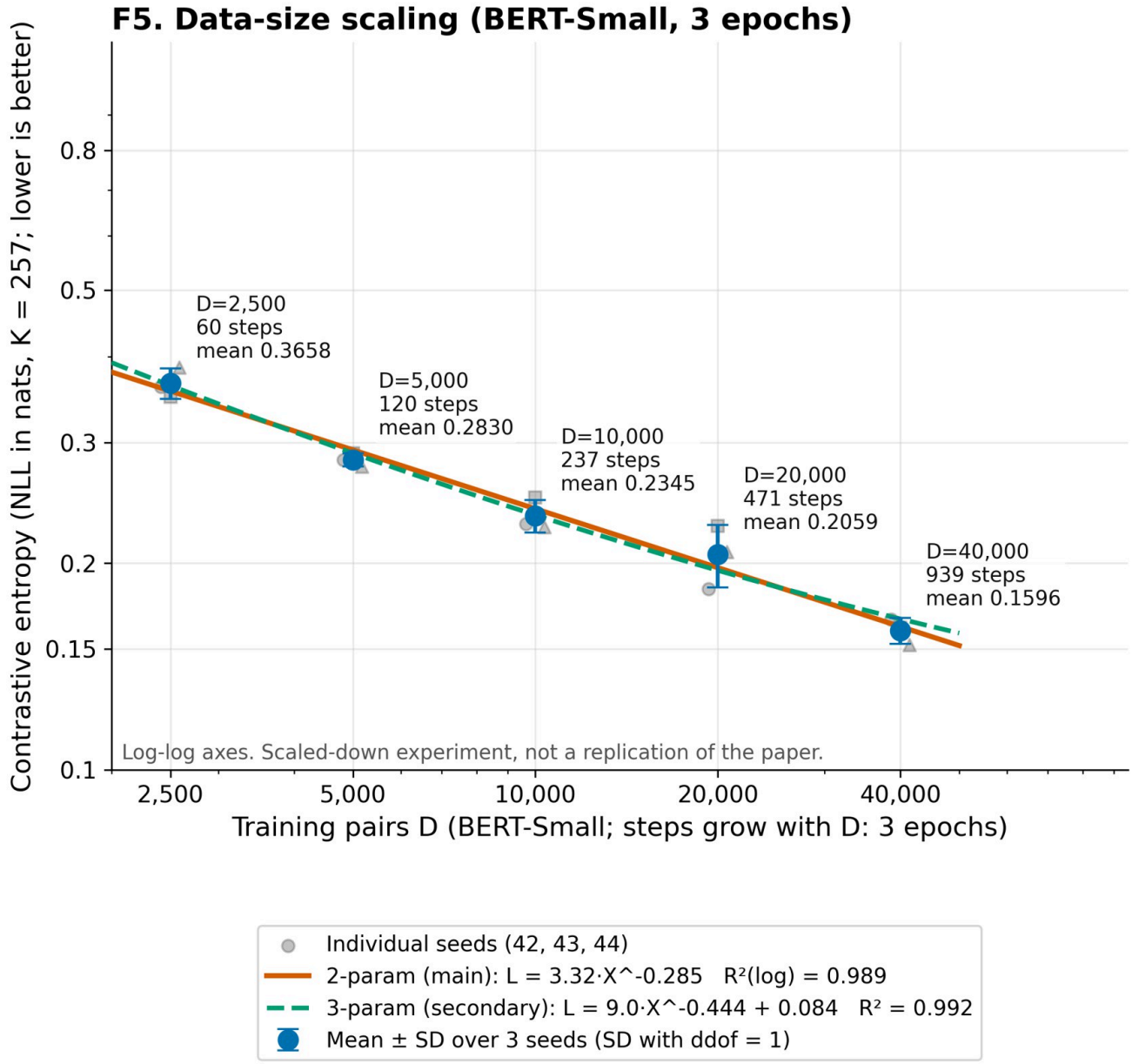


Figure F5. Loss against the number of training pairs for BERT-Small. Note that the number of steps grows with D (60 to 939).

Reading. Loss falls with both N and D. From Tiny to Base the mean loss falls by a factor of 4.6; from D = 2,500 to 40,000 by 2.29. Seed-to-seed SD is between 2.0% and 11.2% of the mean, so the "small spread" claim of the original SUMMARY is too strong for the closest points. All models are far below random scoring (F10).

F6. Residuals of the two fits (error bars = SD of the observed mean, not of the residual)

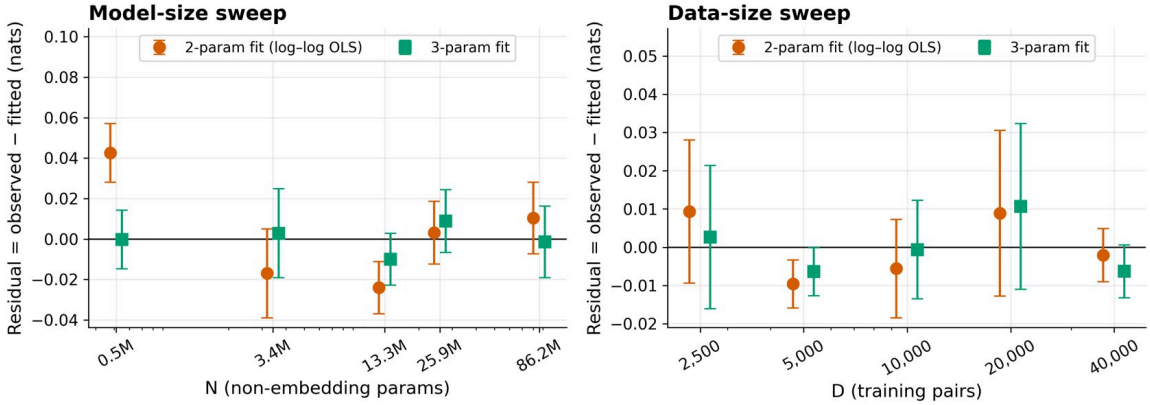
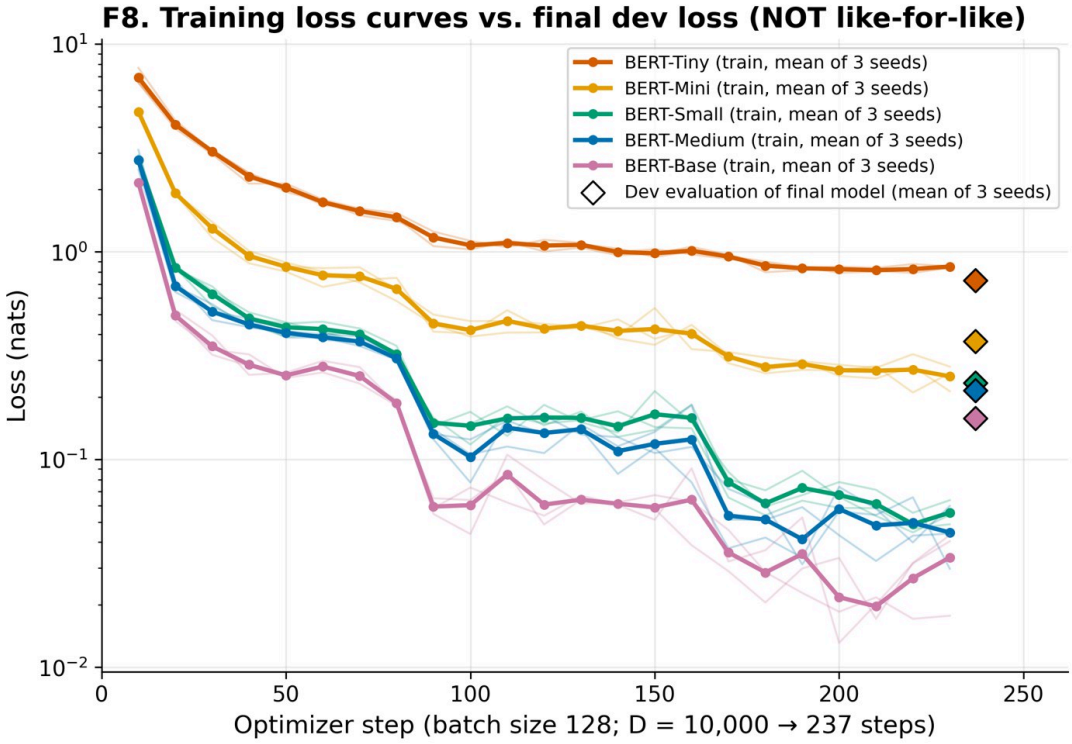


Figure F6. Residuals of the two fits. The 2-parameter fit under-predicts the smallest model; the 3-parameter fit follows all points more closely but has more freedom.

F7. Results table (BERT-Small, D = 10,000 appears in both sweeps; it is one set of 3 runs)

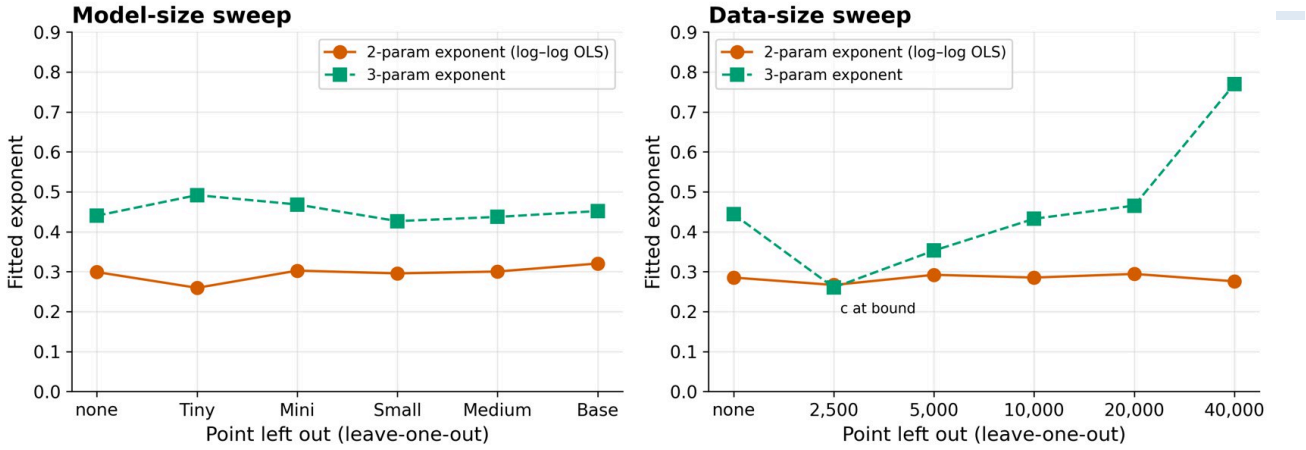
Config	Model	Non-emb. params N	Pairs D	Steps	Seeds	Mean ± SD (nats)	Seed values (42, 43, 44)	Train time (s)
Model sweep	BERT-Tiny	512,128	10,000	237	3	0.7265 ± 0.0145	0.7424, 0.7232, 0.7140	21.3
Model sweep	BERT-Mini	3,422,208	10,000	237	3	0.3707 ± 0.0220	0.3759, 0.3466, 0.3896	34.7
Model sweep	BERT-Small	13,266,176	10,000	237	3	0.2345 ± 0.0129	0.2281, 0.2493, 0.2261	41.4
Model sweep	BERT-Medium	25,875,712	10,000	237	3	0.2149 ± 0.0155	0.1977, 0.2278, 0.2191	75.8
Model sweep	BERT-Base	86,235,648	10,000	237	3	0.1581 ± 0.0177	0.1785, 0.1466, 0.1492	201.2
Data sweep	BERT-Small	13,266,176	2,500	60	3	0.3658 ± 0.0187	0.3618, 0.3494, 0.3862	11.2
Data sweep	BERT-Small	13,266,176	5,000	120	3	0.2830 ± 0.0063	0.2831, 0.2892, 0.2766	21.4
Data sweep	BERT-Small	13,266,176	10,000	237	3	0.2345 ± 0.0129	0.2281, 0.2493, 0.2261	41.4
Data sweep	BERT-Small	13,266,176	20,000	471	3	0.2059 ± 0.0217	0.1834, 0.2266, 0.2078	82.8
Data sweep	BERT-Small	13,266,176	40,000	939	3	0.1596 ± 0.0069	0.1656, 0.1612, 0.1520	165.6

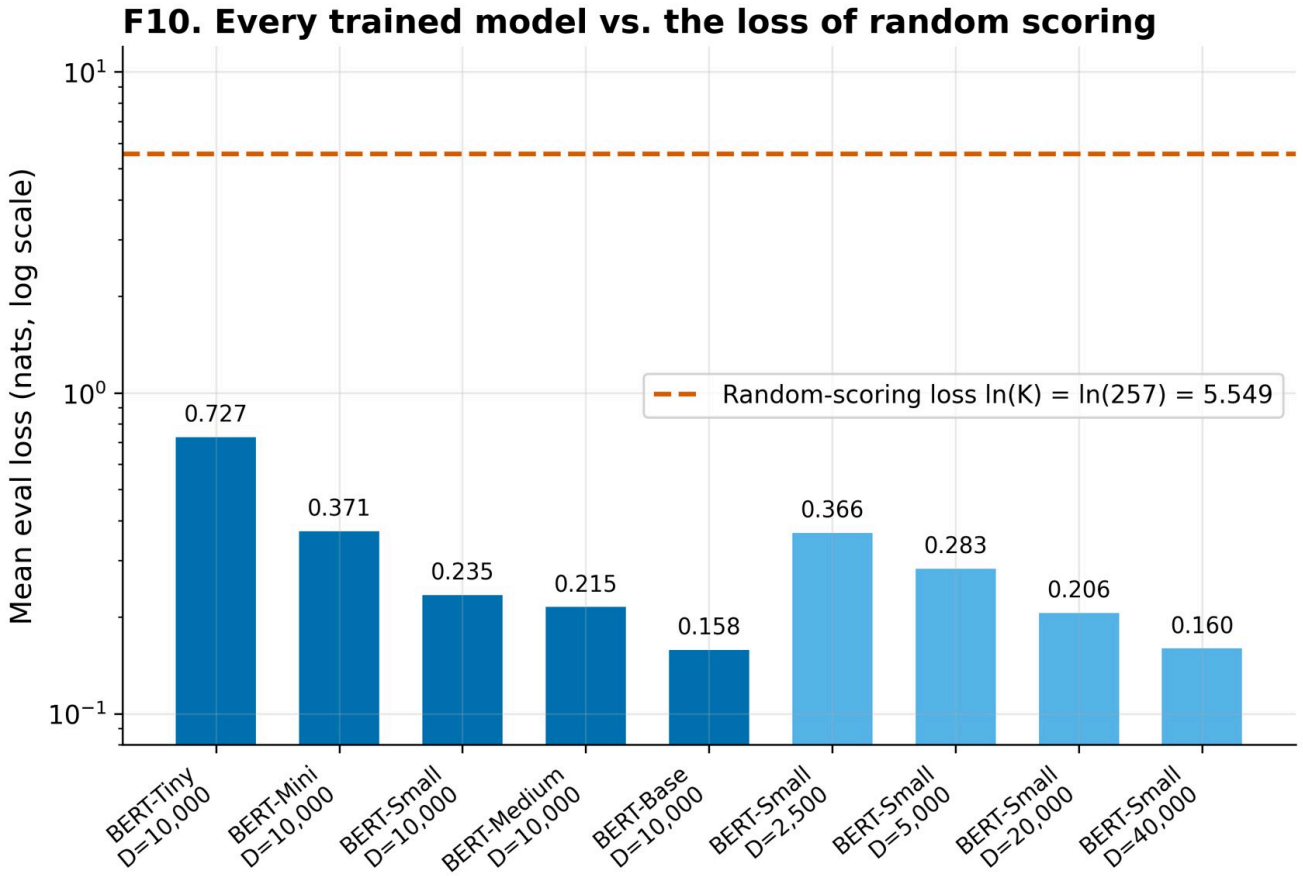


Train curve: mean loss over each 10-step window, dropout on, bf16, on seen pairs, shared random negatives (K = 257).
Diamonds: eval mode, fp32, held-out dev queries, 100k pool. Same K and formula but different mode, data and averaging → no overfitting claim.

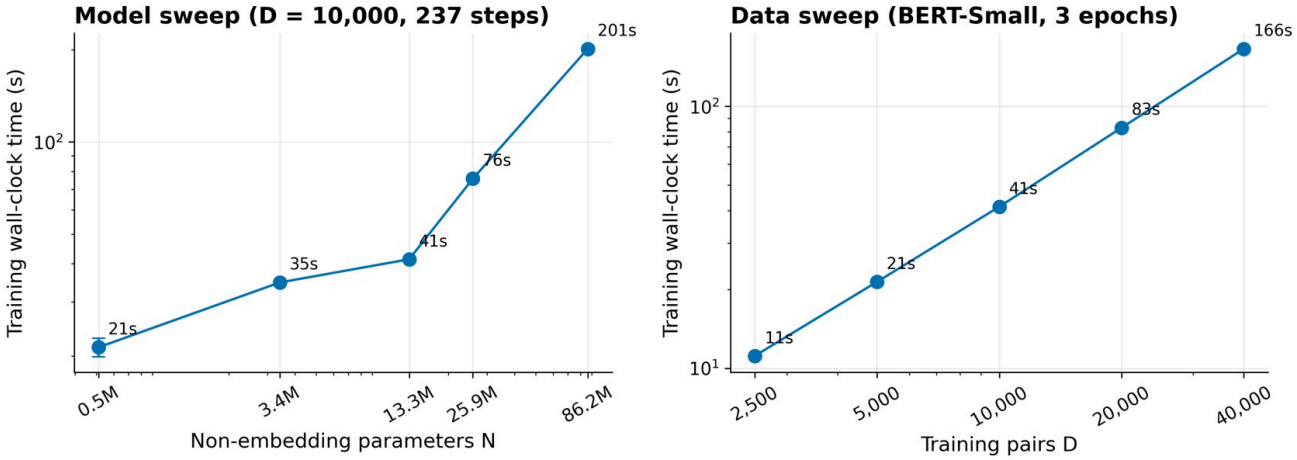
Figure F8. Training loss per 10-step window and the final dev loss (diamonds). Different data, mode and averaging: not comparable, no over-fitting statement.

F9. Stability of the fitted exponent when one of the 5 points is left out

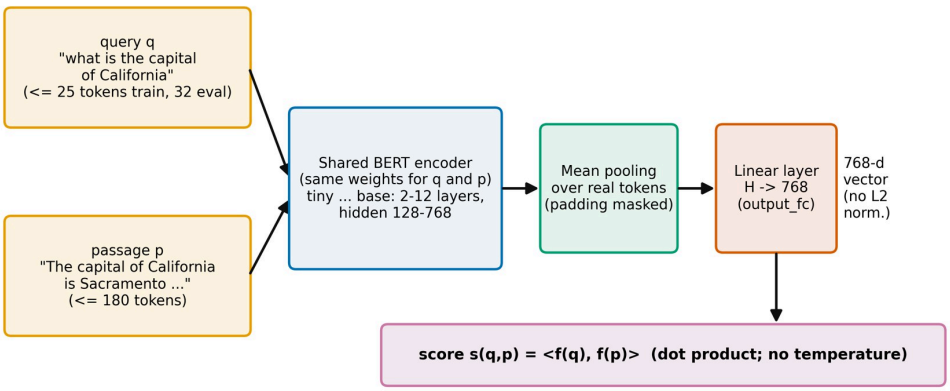




F11. Training time (mean ± SD over 3 seeds; one RTX 5880 Ada GPU, bf16)



F2. DREncoder: one shared encoder for queries and passages



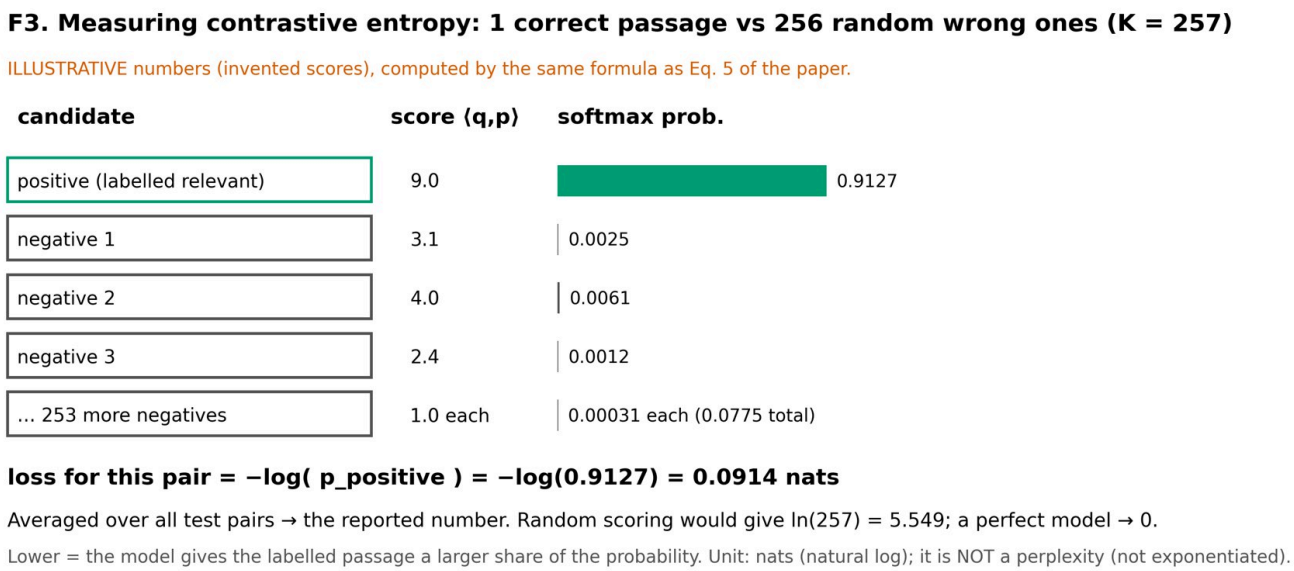


Figure F3 uses invented scores to show the arithmetic: p(positive) = 0.9127 gives a loss of 0.0914 nats.

7. Discussion

Same direction as the paper? Within this small scope, yes in direction: loss decreases as N and D increase (paper: Fig. 3, Fig. 4). Nothing stronger may be said: the exponents, constants and fit qualities are not comparable with Tables 1–2 (Section 5.3), and nothing here tests annotation quality (Sec. 4.3) or budget allocation (Sec. 5).

Differences from the paper. Short schedules (237 steps for the model sweep), final-not-best checkpoint, smaller evaluation, BERT-Small for the data sweep, one dataset. In such short runs larger models learn faster per step (F8), so the model-size curve mixes capacity and speed of learning; the data sweep mixes data and steps.

Limitations. Five points per sweep, three seeds; a 3-parameter fit with 2 degrees of freedom; no test of extrapolation; no hard negatives; corpus truncated; the evaluation set is the 1,000 smallest qids; the learning rate was chosen from two values with an almost untrained model (losses 5.4256 and 4.9708 against chance 5.549).

Threats to validity. *Internal:* training length differs across D; evaluation seed fixed (SD excludes evaluation sampling); early seed-42 runs were produced before the orchestrator was rewritten (their configuration was verified from `training_args.bin`). *External:* one dataset, one model family, in-domain only. *Construct:* contrastive entropy with 256 random negatives measures random-negative discrimination, not ranking quality (the paper also notes the metrics are not equivalent, Sec. 6).

8. How to reproduce

```
# training + evaluation (original orchestrator, background run)
export MSMARCO_DIR=<path to MS MARCO files>
cd drscale_exp && nohup python3 run_all.py > logs/run.log 2>&1 &
# audit and package (from deliverables/scripts)
python3 -I audit_data.py; python3 -I audit_model.py; python3 -I audit_config.py
python3 -I recompute.py; python3 -I make_figures.py; python3 -I build_docs.py; python3 -I check_numbers.py
```

Environment: Python 3.11.7, PyTorch 2.11.0+cu128, Transformers 4.52.4, NumPy 1.26.4, SciPy 1.16.3, pandas 2.3.3, Matplotlib 3.10.6; original repository `jingtaozhan/DRScale`, commit `81453c3b67486641ad282d58f819392a08561e77`. Training is not bit-for-bit deterministic on GPU, so re-running will give slightly different losses.

9. Appendix

A1. Parameter counts (verified by building each model on CPU)

Model	Layers L	Hidden H	Transformer layers only	+ BERT pooler (unused)	= encoder non-emb. ("no fc")	+ output_fc	= non-emb. incl. fc (used on x-axis)	same in units of 2^20	Total params
BERT-Tiny	2	128	396,544	16,512	413,056	99,072	512,128	0.5	4,484,992
BERT-Mini	4	256	3,159,040	65,792	3,224,832	197,376	3,422,208	3.3	11,367,936
BERT-Small	4	512	12,609,536	262,656	12,872,192	393,984	13,266,176	12.7	29,157,632
BERT-Medium	8	512	25,219,072	262,656	25,481,728	393,984	25,875,712	24.7	41,767,168

BERT-Base	12	768	85,054,464	590,592	85,645,056	590,592	86,235,648	82.2	110,072,832

A2. Leave-one-out fits

Sweep	Point left out	2-param exponent	R^2 (log)	3-param exponent	3-param c	bound flag
Model size N	512,128	0.259	0.981	0.492	0.108	—
Model size N	3,422,208	0.303	0.987	0.468	0.106	—
Model size N	13,266,176	0.296	0.994	0.426	0.090	—
Model size N	25,875,712	0.300	0.985	0.437	0.087	—
Model size N	86,235,648	0.320	0.989	0.452	0.101	—
Data size D	2,500	0.267	0.984	0.260	0.000	c at bound
Data size D	5,000	0.292	0.992	0.353	0.042	—
Data size D	10,000	0.285	0.990	0.432	0.081	—
Data size D	20,000	0.294	0.995	0.465	0.083	—
Data size D	40,000	0.276	0.978	0.770	0.166	—

A3. Per-seed fits

Sweep	Seed	2-param exponent	R^2 (log)	3-param exponent	3-param c	R^2 (3-param)	bound flag
model	42	0.292	0.955	0.489	0.115	0.997	—
data	42	0.288	0.983	0.548	0.107	0.998	—
model	43	0.299	0.984	0.469	0.109	0.995	—
data	43	0.258	0.958	0.247	0.000	0.972	c at bound
model	44	0.307	0.987	0.367	0.047	0.996	—
data	44	0.310	0.966	0.669	0.124	0.979	—

A4. Glossary. *NLL / contrastive entropy*: negative log of the softmax probability of the positive. *Nat*: unit of NLL with natural log. *Non-embedding parameters*: parameters excluding word/position embeddings. *SD*: sample standard deviation with $n - 1$. R^2: fraction of variance explained by the fit. *Seed*: number that fixes a random generator. *bf16*: 16-bit floating-point format used during training.

A5. Raw data. `data/results/*.json` (one file per run), `data/results.csv`, `data/results_summary.csv`, `data/figure_data/*.csv`, `data/run.log`, `data/status.json`, `tables/`, `audit/`.

A6. Sources. Paper: arXiv 2403.18684v2 (PDF read locally; text extracted with PyMuPDF). Metadata: Crossref API and arXiv API, fetched 2026-10-29. Ranking: CORE 2023 portal (SIGIR = A*); CCF 2026 list seen via a third-party mirror (official site not accessed). Repository commit 81453c3b67486641ad282d58f819392a08561e77.